

COURSE SYLLABUS

Diplomado en Inteligencia Artificial en Ciencias de la Salud

AI in Clinical Practice: A Unified NIH/AAMC + EU/German Curriculum

Course Information

Course Title	AI in Clinical Practice
Program	Diplomado en Inteligencia Artificial en Ciencias de la Salud
Duration	1 Semester — 25 weeks
Dates	October 1, 2026 – March 24, 2027
Contact Hours	6 hrs/week (3 hrs virtual classroom + 3 hrs Virtual Clinical Sandbox lab)
Total Hours	150 hrs (75 hrs theory + 75 hrs lab, approx.)
Format	Fully virtual — no hospital or clinical-lab access required
Credential Awarded	AI Competency Certificate (NIH/AAMC + German/EU dual-compliant)

Course Instructor

Dr. Saravana Prakash Thirumuruganandham *Educator, STEM Specialist, Professor of Physics, Scientist*

Dr. Thirumuruganandham is a computational physicist and Founder & Director of Quantum Computing at HessQ.Tech, building quantum-inspired technologies for encryption, optimization, and HPC acceleration. He holds a Ph.D. in Physics (*cum laude*) from the Technical University of Kaiserslautern, Germany, with postdoctoral training at Arizona State University (DTRA-funded) and the University of Alabama at Birmingham School of Medicine (NIH-funded). He has authored 40 Scopus-indexed peer-reviewed articles and serves as editor and reviewer for nearly 50 journals, including Topic Editor at *Frontiers in Physics*.

With 14 years of STEM teaching experience across Latin American universities in Spanish, he currently directs R&D at SIT Health & SMARTCO (Quito, Ecuador) and previously held professorships at Universidad Tecnológica Indoamérica and Yachay Tech University.

Career highlights:

- Postdoctoral research at Arizona State University (DTRA-funded) and UAB School of Medicine (NIH-funded)
- Author of 40 Scopus-indexed peer-reviewed publications
- Review editor for nearly 50 peer-reviewed journals
- American Physical Society member; chaired multiple sessions at the 2023 March and April Meetings
- Biophysical Society member (USA)
- Advisory committee member, Ecuador's national supercomputing facility (Quinde-I); doctoral supervisory eligibility, University of A Coruña, Spain

Role in this program: Dr. Thirumuruganandham's background in computational physics and quantum-inspired computing directly supports the mathematical and algorithmic foundations covered throughout this course (Word Error Rate, transformer attention, sensitivity/specificity, explainability methods, differential privacy, and related frameworks in the Theory, Math & Industry Adoption materials).

Program Direction: [Your Name], Director of R&D, SIT Health, serves as Program Director for this Diplomado, overseeing curriculum design, institutional coordination, and alignment with SIT Health & SMARTCO's broader R&D initiatives in Quito, Ecuador.

This course prepares physicians, nurses, and allied health students to safely and effectively use artificial intelligence tools in clinical practice. It integrates two frameworks simultaneously: the **NIH/AAMC Clinical AI Competency Model** (Human-in-the-Loop safety, avoidance of automation bias, multi-modal data integration) and the **EU AI Act & German Digital Health Standards** (GDPR data protection, the DiGA pathway, and German quality pillars of Functionality, Robustness, and Comprehensibility).

Because this cohort does not have hospital or clinical-lab access, all hands-on work takes place in a **Virtual Clinical Sandbox (VCS)** using synthetic patient records (Synthea), de-identified public datasets (MIMIC-CXR, MIMIC-IV), sandbox EHR systems (OpenEMR/GNU Health), and standardized-patient roleplay. No real patient data is used at any point in this course.

Learning Objectives

By the end of this course, students will be able to:

1. Apply the Human-in-the-Loop (HITL) principle to verify AI-generated clinical documentation, diagnostic recommendations, and care plans.
2. Identify and manually override incorrect or biased AI diagnostic output, with proper documentation.
3. Explain GDPR-driven data-sovereignty requirements and correctly anonymize data for secondary research use.

4. Map liability, classify tools under EU AI Act risk tiers, and disclose AI involvement to patients appropriately.
5. Integrate all of the above into a single, independently-run simulated clinical shift.

Required Materials & Tools

- Reliable internet connection and a webcam/microphone for virtual sessions and roleplay
- Institutional accounts (provided): Synthea, sandbox EHR (OpenEMR/GNU Health), ambient-scribe sandbox license, virtual OSCE platform
- No textbook purchase required — all readings are open-access and linked in the weekly schedule below
- **Data safety policy:** students may never upload real patient information to any tool used in this course; only synthetic or fully de-identified data is permitted

Grading & Assessment

Component	Weight	Description
Weekly Discussion/Reflection Questions	20%	USMLE-norm question + medical-education reflection question, each week
Monthly Competency Checkoffs	40%	Graded pass/fail gates at the end of Chapters 1–4 (Weeks 4, 8, 12, 17)
Mock OSCE (Consolidation)	15%	Weeks 19–20
Capstone Shifts (×2)	20%	Weeks 22–23, graded for consistency across both shifts
Final Certification Checkoff	5%	Week 25

Passing standard: Students must pass each monthly checkoff to advance to the next chapter. A failed checkoff requires a documented repeat of that block’s Virtual Clinical Sandbox exercise before re-attempting.

Weekly sufficiency check: At the end of each week, a student has demonstrated sufficient knowledge transfer for that unit if they can (a) answer the week’s USMLE-norm question correctly, and (b) explain the week’s clinical-practice advantage unprompted, in their own words.

Weekly Topic & Reading Schedule

Required

Wk	Dates	Topic	Reading / Reference	Assessment
1	Oct 1-7, 2026	Introduction to AI in Medicine	NIH/AAMC Clinical AI Competency Model; HITL workflow overview	Discussion Q
2	Oct 8-14, 2026	Prompt Engineering — The Four Rs	AAFP “Four Rs” framework; prompt literacy in clinical training	Discussion Q
3	Oct 15-21, 2026	Ambient Scribing in Practice	German GP digital-literacy gap report; clinical AI evidence base	Discussion Q
4	Oct 22-28, 2026	Synthesis & Checkoff	Ambient-scribing outcomes review	Checkoff 1 (graded)
5	Oct 29-Nov 4, 2026	Critical Appraisal of AI Diagnostics	AI board-exam vs. bedside-reasoning dissociation studies	Discussion Q
6	Nov 5-11, 2026	German Quality Pillars	German quality-pillars framework; GDPR/XAI compliance	Discussion Q
7	Nov 12-18, 2026	The Sandbox Override Exercise	Clinical AI bias case studies	Discussion Q
8	Nov 19-25, 2026	Synthesis & Checkoff	—	Checkoff 2 (graded)
9	Nov 26-Dec 2, 2026	AI-Enhanced Care Planning	AI-enhanced clinical encounters literature	Discussion Q
10	Dec 3-9, 2026	GDPR & Data Sovereignty	EU GDPR healthcare compliance guide; sovereign-hosting rationale	Discussion Q
11	Dec 10-16, 2026	Anonymization & Secondary Data Use	DiGA (Digital Health Applications) pathway documentation	Discussion Q

12	Dec 17-23, 2026	Synthesis & Checkoff	—	Checkoff 3 (graded)
—	Dec 24-Jan 6	<i>Recommended institutional recess</i>	—	—
13	Dec 24-30, 2026	Liability Mapping	NIH/AAMC liability & practice-based learning guidance	Discussion Q
14	Dec 31-Jan 6, 2027	EU AI Act Risk Tiers	EU AI Act overview; MDCG 2025-6 guidance	Discussion Q
15	Jan 7-13, 2027	Informed Consent & Transparency	EU AI Act & patient-consent literature	Discussion Q
16	Jan 14-20, 2027	Capstone Preparation	—	Ungraded dry run
17	Jan 21-27, 2027	Final Checkoff	—	Checkoff 4 (graded)
18	Jan 28-Feb 3, 2027	Integration Station Design	—	Peer station design
19	Feb 4-10, 2027	Mock OSCE — Round 1	—	OSCE Round 1 (graded)
20	Feb 11-17, 2027	Mock OSCE — Round 2 + Remediation	—	OSCE Round 2 (graded)
21	Feb 18-24, 2027	Consolidation Wrap-Up	—	Readiness sign-off
22	Feb 25-Mar 3, 2027	Capstone Shift 1	—	Capstone Shift 1 (graded)
23	Mar 4-10, 2027	Capstone Shift 2	—	Capstone Shift 2 (graded)
24	Mar 11-17, 2027	Remediation / Retake Window	—	As needed
25	Mar 18-24, 2027	Certification & Course Close	—	Final Checkoff (graded)

Full reference URLs are listed in the Reference Library appendix below.

Course Policies

Data safety: This is a zero-real-PHI course. All Virtual Clinical Sandbox activities use synthetic (Synthea) or fully de-identified public data (MIMIC-CXR/MIMIC-IV). Uploading real patient information to any tool in this course is a policy violation.

Academic integrity: Discussion questions and reflection prompts are meant to be answered individually unless a session is explicitly designated group work. Checkoffs are individually graded and proctored.

Attendance: Each week’s 3-hour virtual session and 3-hour Virtual Clinical Sandbox block are both required. Students who miss a checkoff week must arrange a make-up session with the instructor before the next chapter begins.

Remediation policy: A failed monthly checkoff or capstone shift is not a course failure — it triggers a targeted, narrowly-scoped remediation (see Week 24 model) focused only on the specific named gap, followed by a re-attempt.

Accommodations: Students requiring accommodations should contact the instructor during Week 1 to arrange adjustments to session format, timing, or assessment delivery.

Appendix: Reference Library

#	Source	Topic
1	NIH/AAMC Clinical AI Competency Model (pubmed.ncbi.nlm.nih.gov/36731054)	Overall competency framework
2	German AI Healthcare Compliance Guide 2026 (germancomplianceinstitute.com)	EU AI Act & GDPR
3	Clinical AI evidence base (pmc.ncbi.nlm.nih.gov/articles/PMC11993958)	Bias, liability, evidence review
4	German quality pillars overview	Functionality/Robustness/Comprehensibility
5	Prompt literacy in clinical training (pmc.ncbi.nlm.nih.gov/articles/PMC11822726)	Prompt engineering
6	Human-in-the-Loop workflow (pubmed.ncbi.nlm.nih.gov/42710030)	HITL foundations
7	Clinical AI evaluation methods (pmc.ncbi.nlm.nih.gov/articles/PMC12820482)	Diagnostic appraisal
8	AAFP “Four Rs” prompting framework (aafp.org/fpm/2025/0900)	Prompt engineering
9	GDPR / Explainable AI (pmc.ncbi.nlm.nih.gov/articles/PMC12978887)	XAI compliance
10	Physician AI Handbook — German quality pillars	Quality pillars
11	Local/open-weight deployment rationale (VDE proceedings)	Data sovereignty
12	AI board-exam vs. bedside reasoning (news.syenza.com)	Diagnostic dissociation
13	AI-enhanced clinical encounters (pmc.ncbi.nlm.nih.gov/articles/PMC7980122)	Care planning
14	Final exam/capstone checklist methodology (pubmed.ncbi.nlm.nih.gov/35612209)	Capstone design

15	DiGA pathway (pmc.ncbi.nlm.nih.gov/articles/PMC12973773)	Digital Health Applications
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Instructors should re-verify all open-access links before distribution each term, as URLs and article availability can change.

Companion Teaching Materials

This syllabus is designed to be used alongside four companion decks already prepared for this course: 1. **Teaching & Instructional Material Deck** (31 slides) — full teaching script and VCS activity instructions per unit 2. **Time, Training & Practice Deck** (25 slides) — time-saved data, AI-training background, and practice questions per unit 3. **Theory, Math & Industry Adoption Deck** (25 slides) — equations, algorithms, company adoption, and three-part assessment per unit 4. **Weekly Agenda & Time Breakdown Deck** (25 slides) — minute-by-minute class schedule per unit, in both minutes and decimal hours